

EC3213 — Money, Credit and Banking

Winter 2025 Examination · Paper 1 · Full Marking Scheme

Module	EC3213 — Money, Credit and Banking
Internal Examiner	Dr J. Gao
Institution	University College Cork
Session	Winter 2025, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions.

Own-figure rule. Where a question part carries forward from an earlier numerical step (e.g. A1(a)'s arbitrage direction feeding into its profit calculation), an answer that applies the correct method consistently from an earlier error still earns the marks available for the later steps.

Source basis. This scheme follows Dr Gao's own EC3213 lecture material closely — several sub-questions (A1(b), A2(a), A3(c)) are drawn essentially verbatim from the module's own review-question sets, and the worked examples (the money multiplier's €1,000→€10,000 case, the Taylor rule numerical example) are the lecture's own examples, reproduced here with independently verified arithmetic. A3(b)'s comparison of money demand theories follows the standard treatment (Mishkin) for Friedman's modern quantity theory specifically, since the lecture slides referenced it by name without reproducing the full theory — flagged at that question.

Section A · Question 1

Covered interest arbitrage, CIP/PPP interrelationships, and the monetary policy chain —
100 marks

A1(a)

[40 marks]

You have been given the following information: $r_{\$} = 4\%$ (annual, 3-month US-dollar commercial paper), $r_{£} = 6\%$ (annual, 3-month British-pound commercial paper), $S(\$/£) = 1.5000$, $F_{1/4}(\$/£) = 1.4988$. On the basis of the precise criteria: (i) In which commercial paper would you invest? (ii) In which currency would you borrow? (iii) How would you arbitrage? (iv) What is the profit from interest arbitrage per dollar borrowed?

A1(b)

[30 marks]

Suppose that real interest rates are equal for all countries in the world. Does this imply anything for the relationship between covered interest-rate parity and the PPP condition? Why?

A1(c)

[30 marks]

With the aid of a diagram, explain the linkages between Central Bank Tools, Policy Instruments, Intermediate Targets and Goals of Monetary Policy.

FULL MODEL ANSWER

Setting up the comparison: with a 3-month ($T = 0.25$ year) horizon, the two options are: (A) invest \$1 in US commercial paper at 4% p.a., uncovered; or (B) convert \$1 to £ at spot, invest in British-pound commercial paper at 6% p.a., and sell the £ proceeds forward at $F_{1/4}$ to lock in the \$ conversion — the **covered** alternative. Comparing the two tells us immediately whether arbitrage is available, since covered interest parity (CIP) should otherwise equalise the two returns.

Strategy	Calculation	Result per \$1
(A) Invest \$1 in US CP	$1 \times (1 + 0.04 \times 0.25)$	\$1.010000
Convert \$1 to £ at spot	$1 \div 1.5000$	£0.666667
Invest £0.666667 in UK CP	$0.666667 \times (1 + 0.06 \times 0.25)$	£0.676667
(B) Convert back at $F_{1/4} = 1.4988$	0.676667×1.4988	\$1.014188

(i) Which commercial paper to invest in: the covered British-pound strategy (B) returns **\$1.014188** per dollar, strictly more than the US strategy (A)'s \$1.010000 — so invest in **British-pound commercial paper**, covered forward.

(ii) Which currency to borrow: since the profitable side of the trade is investing in £ CP, the arbitrage is funded by **borrowing in US dollars** at 4% — borrowing in the currency you are *not* investing in is what makes this a self-financing (zero-net-investment) arbitrage rather than simply a directional bet.

(iii) How to arbitrage: borrow \$1 (or any amount) at $r\$ = 4\%$; convert immediately to £ at the spot rate $S = 1.5000$; invest the £ proceeds in British-pound commercial paper at $r£ = 6\%$; simultaneously sell the maturing £ proceeds forward 3 months at $F_{1/4} = 1.4988$ to lock in the \$ conversion today; at maturity, convert the £ proceeds back to \$ at the contracted forward rate, repay the \$ loan plus interest, and keep the difference as arbitrage profit — with every leg of the trade covered by the forward contract, there is no exchange-rate risk in the strategy at all.

(iv) Profit from interest arbitrage per dollar borrowed: $\$1.014188$ (covered £ strategy) — $\$1.010000$ (cost of repaying the \$ loan) = **\$0.004188 profit per dollar borrowed**, or 0.4188% over the 3-month period.

EXAM TECHNIQUE

A quick sanity check: since $r£ (6\%)$ exceeds $r\$ (4\%)$, CIP predicts the £ should trade at a forward *discount* against the \$ (F should be below $S = 1.5000$ — higher-interest currencies depreciate forward under CIP to prevent arbitrage). The CIP-consistent forward would be $F = S \times (1+r\$T)/(1+r£T) = 1.5000 \times 1.01/1.015 \approx \mathbf{1.4926}$. The actual quoted forward (1.4988) is *higher* than this — the £ hasn't depreciated forward as much as CIP requires — which is exactly why covered investment in £ CP is the profitable side of the trade. If the actual forward had instead been quoted *below* 1.4926, the arbitrage would run the other way (invest in \$, borrow £).

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly computes both the uncovered US return and the covered £ return AND correctly identifies which CP to invest in and which currency to borrow, consistent with the calculation AND correctly describes the arbitrage mechanics AND arrives at \$0.004188 (or a consistent own-figure result).
21-30	21-30	Both returns correctly calculated and the direction of arbitrage correctly identified, but the arbitrage mechanics description is incomplete or the profit figure has an arithmetic error.
11-20	11-20	Only one of the two strategies' returns correctly calculated, or the direction of arbitrage (which currency to borrow vs. invest) is reversed.
0-10	0-10	No coherent covered-interest-arbitrage calculation attempted.

Yes — equal real interest rates is precisely the condition that connects covered interest parity to PPP: the module's own "interrelationship of the parity conditions" result shows that **uncovered** interest parity (UIP), the expectations form of PPP, and the **Fisher-open condition** (equal expected real interest rates across countries) are mutually derivable — any two of the three imply the third.

Tracing the logic through to CIP specifically: CIP itself is a pure, risk-free no-arbitrage condition on financial assets and holds essentially by construction (as in Part (a)) — it doesn't reference expectations at all. But under UIP (the forward rate is, in expectation, an unbiased predictor of the future spot rate — i.e. no risk premium), CIP and UIP coincide numerically: the forward premium implied by CIP *equals* the expected rate of currency depreciation. If, in addition, real interest rates are equal across countries (the Fisher-open condition), then by the Fisher effect operating separately within each country, the nominal interest differential between the two countries must equal purely the *expected inflation differential* between them — nothing else, since the real-rate components cancel out. Combining the two: the CIP/UIP-implied forward premium (= nominal interest differential, when UIP holds) equals the expected inflation differential, which is exactly what the **expectations form of PPP** predicts for expected currency depreciation.

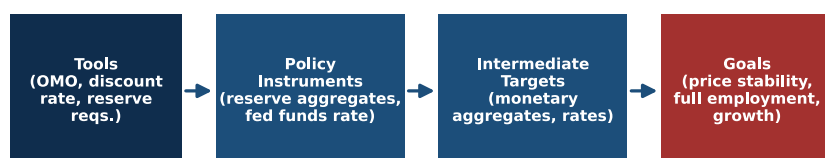
The upshot: equal real interest rates across countries is the condition that makes a covered financial-market relationship (CIP, combined with UIP holding) **consistent with, and effectively equivalent to**, the goods-market PPP condition holding in its expectations form. Without real-rate equality, the nominal interest differential CIP prices into the forward rate could reflect differences in real returns as much as differences in expected inflation, breaking the clean link to PPP.

Mark Allocation — Part (b) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	Correctly identifies that UIP, expectations-PPP, and the Fisher-open condition are mutually derivable AND correctly traces the Fisher-effect logic showing equal real rates implies the nominal differential equals the expected inflation differential AND correctly connects this to CIP's forward premium matching PPP's predicted depreciation.
15-22	15-22	Correctly states that equal real rates connects interest parity to PPP via the Fisher effect, but the specific mechanism (nominal differential = inflation differential = forward premium = expected depreciation) is not fully traced through.
8-14	8-14	Vague reference to "Fisher effect" or "the conditions are related" without a clear mechanism.
0-7	0-7	No credible answer given, or CIP and UIP are conflated without acknowledgement of the distinction.

The chain: a central bank cannot directly control its ultimate goals (price stability, full employment, sustainable growth) — it can only pull the levers genuinely under its control and rely on a chain of fairly reliable, but imperfect, links to eventually influence those goals.

Linkages Between Central Bank Tools, Instruments, Targets and Goals



Each link is chosen for observability, measurability, controllability, and predictable effect on the next link

Fig. 1 — The chain from central bank tools through policy instruments and intermediate targets to the ultimate goals of monetary policy.

- **Tools:** what the central bank directly controls — open market operations, the discount rate, and reserve requirements (see Q. A2(a) and A3(c)).
- **Policy instruments (operating targets):** variables the tools directly and immediately move — reserve aggregates or the federal funds rate itself — chosen because they are observable, measurable, and controllable on a near-daily basis.
- **Intermediate targets:** variables one step further removed — monetary aggregates (M1, M2) or longer-term interest rates — that the central bank can influence fairly reliably via the policy instrument, and which in turn have a more predictable relationship with the ultimate goals than the tools do directly.
- **Goals:** the ultimate objectives — price stability, maximum sustainable employment, and moderate long-term interest rates/growth — that monetary policy exists to serve, but which the central bank can only reach indirectly, with long and variable lags, through the chain above.

Why the chain is structured this way: each link is selected specifically for **observability and measurability, controllability**, and a **predictable effect on the next link in the chain** — a central bank targets intermediate variables rather than jumping straight to its goals precisely because goals like inflation and employment are observed only with a lag and respond to policy only after a long and variable delay, whereas reserve aggregates or the fed funds rate can be observed and adjusted almost immediately.

Mark Allocation — Part (c) (30 marks)

Tier	Marks	What separates this tier
23-30	23-30	All four links in the chain correctly named with a correct example of each AND the diagram correctly drawn showing the sequential linkage AND the observability/controllability/predictability rationale for the chain's structure is explained.
15-22	15-22	All four links correctly named with examples, but the rationale for why the chain is structured this way (rather than targeting goals directly) is not explained.
8-14	8-14	Only two or three links correctly named, or named without examples.
0-7	0-7	No credible chain given.

Question A1 Total: 100 Marks

Section A · Question 2

Open market operations and the fed funds rate, PPP exchange rates, and the money multiplier — 100 marks

A2(a)**[40 marks]**

Explain, with the aid of a diagram, how the Federal Funds interest rate is affected by an Open Market Sale by the Federal Reserve Bank. Illustrate how Open Market Operations are an effective tool for targeting the Fed Funds Rate.

A2(b)**[40 marks]**

Assume that the prices of a standard basket of goods and services in different countries are as follows: United States \$300, Canada C\$450, United Kingdom £180, Japan ¥50,000. (i) From a US perspective, what are the implied PPP exchange rates? (ii) How would the presence of a high sales tax in the United Kingdom, such as the VAT, influence your guess of where the actual value of $S(\$/\pounds)$ would be in comparison with the PPP value of $S(\$/\pounds)$? (iii) Why might there be departures from PPP even if the law of one price holds for every commodity?

A2(c)**[20 marks]**

With the aid of an example, explain the money multiplier.

FULL MODEL ANSWER

The market for reserves: the federal funds rate is the price that clears the **market for bank reserves** — determined by the demand banks have for holding reserves (partly to meet requirements, partly as a cushion against unexpected deposit outflows) and the supply of reserves the Fed makes available.

The demand curve: downward-sloping in the fed funds rate — as i_{ff} falls, the opportunity cost of holding excess reserves (relative to lending them out) falls, so banks want to hold more of them. Since 2008 the Fed has paid interest on reserves (i_{or}) at a fixed spread below the fed funds target, so the demand curve **flattens out (becomes infinitely elastic) once i_{ff} falls to i_{or}** — below that point, banks would rather hold reserves than lend at a rate no better than what the Fed itself pays.

The supply curve: made up of non-borrowed reserves (supplied via OMOs) plus borrowed reserves (banks borrowing directly from the Fed at the discount rate, i_d). If $i_{ff} < i_d$, no bank will borrow from the Fed when cheaper funding is available in the interbank market, so borrowed reserves are zero and supply is **vertical**. If i_{ff} rises above i_d , banks profitably borrow from the Fed and re-lend at i_{ff} , so supply becomes **horizontal (perfectly elastic) at i_d** .

The Market for Reserves: Effect of an Open Market Sale

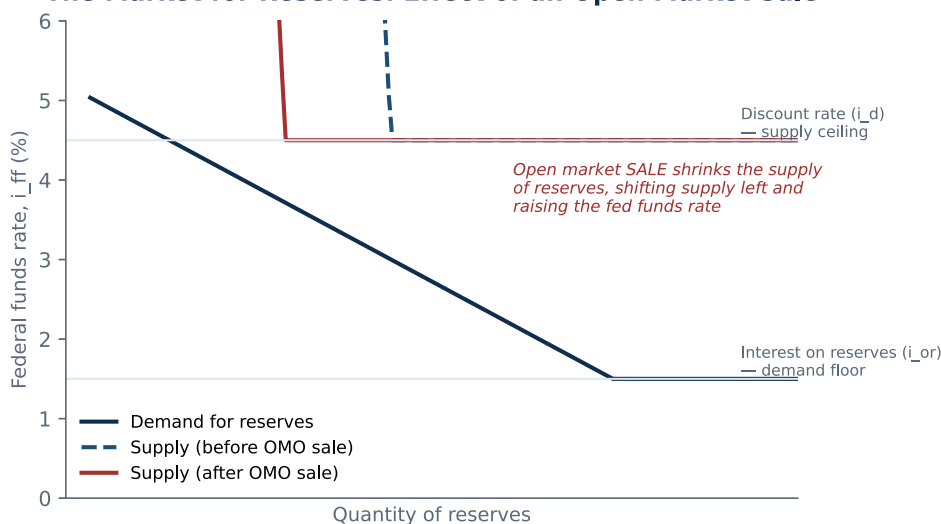


Fig. 1 — An open market sale shrinks the supply of reserves, shifting the (vertical section of the) supply curve left. Where the new equilibrium remains on the downward-sloped section of the demand curve, this raises the federal funds rate from A to B.

Effect of an open market sale: an OMO sale drains reserves from the banking system, shifting the vertical portion of the supply curve to the **left**. Provided the new intersection with demand still falls on the downward-sloped section of the demand curve (i.e. the fed funds rate is still above i_{or}), this **raises the federal funds rate**. Conversely, an open market purchase shifts supply right and lowers i_{ff} .

Why OMOs are an effective targeting tool: OMOs give the Fed **complete control over the volume** transacted (unlike discount lending, where the bank itself decides whether to borrow), are **flexible and precise** — any desired quantity of reserves can be added or drained — and can be **quickly implemented and easily reversed** if conditions change, which is precisely what makes OMOs the Fed's primary conventional tool rather than the discount rate or reserve requirements (see Q. A3(c)).

EXAM TECHNIQUE

A common exam slip is describing the effect of an OMO sale without specifying *where* the new equilibrium falls relative to i_{or} — if reserves remain abundant enough that the intersection stays on the flat section of the demand curve at i_{or} , an OMO sale of typical size has **no effect** on the fed funds rate at all. The diagram's shape, not just the direction of the shift, is what determines the answer.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Demand and supply curves in the market for reserves correctly shaped (demand downward-sloping then flat at i_{or} ; supply vertical then flat at i_d) AND an OMO sale correctly shown shifting supply left, raising i_{ff} on the relevant section AND at least two specific reasons given for why OMOs are an effective targeting tool.
21-30	21-30	Correct diagram shape and correct direction of the OMO sale's effect, but the reasons OMOs are an effective tool are generic or missing.
11-20	11-20	Diagram shape roughly correct but the flat sections (at i_{or} and i_d) are missing or mislabelled, or the direction of the OMO sale's effect is reversed.
0-10	0-10	No credible diagram or explanation given.

(i) Implied PPP exchange rates from a US perspective: absolute PPP says

the same basket should cost the same amount everywhere once converted to a common currency — so the implied PPP rate is simply the ratio of the two countries' basket prices.

Country pair	Calculation	Implied PPP rate
Canada	C\$450 ÷ \$300	C\$1.50 per US\$
United Kingdom	£180 ÷ \$300	£0.60 per US\$ (equivalently, \$1.6667 per £)
Japan	¥50,000 ÷ \$300	¥166.67 per US\$

(ii) The effect of a high UK sales tax (VAT) on actual vs. PPP S(\$/£): VAT is a **domestic consumption tax** — it is charged on goods sold to UK consumers, but goods **exported** from the UK are zero-rated or rebated for VAT, so a foreign buyer effectively pays the *pre-VAT* price, not the £180 domestic consumer-facing price used above. This means the £180 figure **overstates** the price level actually relevant to international trade and arbitrage in this basket's goods — the true, export-relevant UK price is lower than £180. Using this VAT-inflated domestic price to compute the naive PPP rate (\$1.6667 per £) therefore **understates** the pound's true purchasing power in tradeable goods. The actual market exchange rate $S(\$/\text{£})$, which is set by arbitrage in the (VAT-free) traded-goods price level, would be expected to show a **stronger pound** — more dollars per pound — than the naive \$1.6667 PPP estimate calculated from tax-inclusive domestic prices.

(iii) Why PPP can still fail even if the law of one price holds for every commodity: the law of one price applies to *individual* goods; PPP is a statement about *price indices* — baskets of many goods combined using country-specific weights. Even if every single good obeys the law of one price perfectly, the overall basket comparison can still diverge because different countries weight the same goods differently in their consumption baskets, reflecting different tastes and needs. A textbook illustration: people in cold, northern countries consume relatively more heating oil and less olive oil than people in temperate countries. If heating-oil prices rise more than olive-oil prices (with each individually still obeying the law of one price across countries), the heating-oil-heavy country's overall price index rises by more than the olive-oil-heavy country's — a PPP departure driven purely by different basket *weights*, not by any breakdown of the law of one price itself. A second, related reason is that real-world consumption baskets include **non-traded goods and services** (haircuts, local housing, restaurant meals) that are outside the scope of the law of one price altogether, since they cannot be arbitrated across borders.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	All three PPP rates correctly calculated AND the VAT/export-rebate mechanism correctly explained with the correct directional conclusion (actual S stronger than naive PPP) AND the basket-weighting explanation for PPP departures correctly given with a concrete example.
21-30	21-30	PPP rates correctly calculated and the VAT point given, but the direction of the effect on actual vs. PPP $S(\$/\text{£})$ is reversed or unclear, or the basket-weighting explanation for part (iii) is vague.
11-20	11-20	PPP rates correctly calculated, but parts (ii) and (iii) are weak or missing.
0-10	0-10	PPP rates incorrectly calculated, or no credible explanations given for (ii)/(iii).

The money multiplier: the ratio by which the money supply expands relative to an initial injection of reserves into the banking system, since banks re-lend most of any new deposit rather than holding it all as reserves, and each re-lent amount becomes a new deposit at another bank, which again re-lends most of it, and so on.

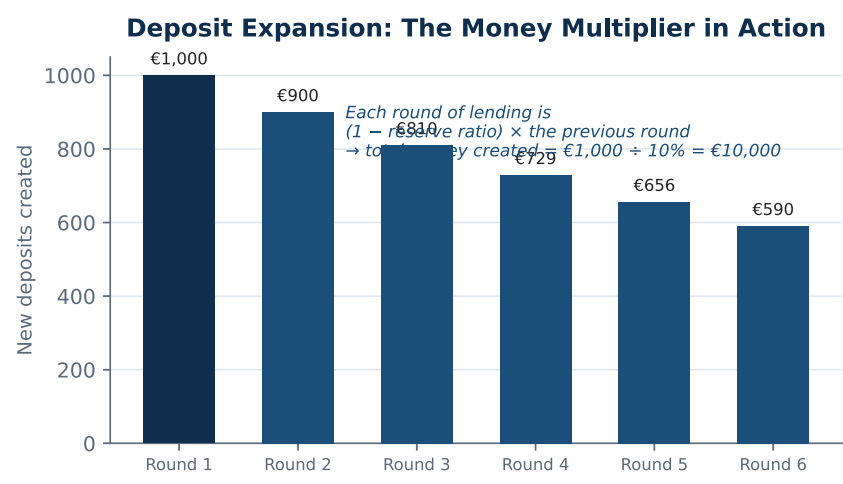


Fig. 2 — Successive rounds of deposit expansion (illustrative: €1,000 initial deposit, 10% reserve ratio), shrinking geometrically and summing to the total money created.

Round	New deposit created (€)	Held as reserves (€)	Re-lent (€)
1	1,000.00	100.00	900.00
2	900.00	90.00	810.00
3	810.00	81.00	729.00
...	(geometric series continues)		
Total money created	€10,000.00		= €1,000 ÷ 10%

Result: with a reserve ratio of 10%, the simple money multiplier is $1 \div 10\% = 10$, so an initial €1,000 deposit ultimately supports €10,000 of money in the banking system — the course's own worked example. In practice, the actual (M1) multiplier is smaller than this simple reserve-ratio formula suggests, since it also depends on the public's currency-to-deposit ratio and banks' holding of excess reserves beyond the legal minimum.

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Correctly explains that banks re-lend deposits net of reserves, repeating across successive rounds AND gives a correct worked numeric example with the multiplier formula (1/reserve ratio).
10-14	10-14	Correctly explains one round of lending increasing the money supply, but does not extend the reasoning to successive rounds or a full worked example.
5-9	5-9	Vague reference to "banks lending out deposits" without a reserve ratio or multiplier formula.
0-4	0-4	No credible explanation given.

Section A · Question 3

The Phillips curve, the Taylor rule, theories of money demand, and monetary policy tools — 100 marks

A3(a)

[30 marks]

Outline the economic reasoning behind: (i) The Accelerationist Phillips Curve; and (ii) The Taylor Rule.

A3(b)

[50 marks]

Compare the Classical Theory of Money Demand, the Keynesian Theory of Money Demand and Milton Friedman's Theory of Money Demand. In your opinion, which theory (theories) of money demand is most appropriate to describe the current economic environment? Why?

A3(c)

[20 marks]

Briefly outline the difference between conventional and unconventional monetary policy tools.

FULL MODEL ANSWER

(i) The Accelerationist Phillips Curve — the reasoning: Phillips's original (1958) finding was a negative statistical relationship between unemployment and (wage) inflation — tight labour markets meant higher wage inflation, and this looked like a stable policy trade-off a central bank could exploit. Friedman's 1968 critique reasoned that workers actually bargain over **real**, not nominal, wages: in a tight labour market they'd demand nominal wage growth in excess of expected price inflation, and if that expected inflation is **adaptive** — this year's expectation is simply last year's actual inflation ($\pi^e = \pi^{t-1}$) — the expectations-augmented Phillips curve collapses to $\pi_t = \pi_{t-1} - \gamma(U_t - U^*)$: it is the *change* in inflation, not its level, that relates to the unemployment gap.

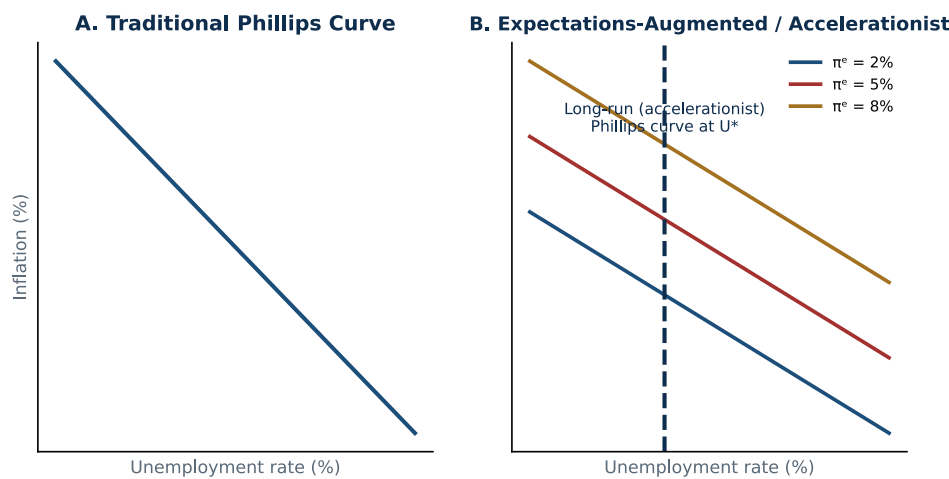


Fig. 1 — The traditional Phillips curve (A) suggests a stable trade-off; the expectations-augmented/accelerationist view (B) shows the curve shifting outward as expected inflation rises, with only a vertical long-run curve at the natural rate U^* .

Why "accelerationist": the model implies a central bank *can* hold unemployment below its natural rate U^* — but only by accepting **ever-accelerating** inflation, since each period's attempt to exploit the trade-off pushes expected inflation up further, shifting the short-run curve outward again the following period. There is no *level* of inflation consistent with U below U^* forever — only a permanently rising one, which is exactly what happened in the US through the 1970s (rising inflation *and* rising unemployment together — stagflation — once the public's expectations caught up with reality), and is widely seen as vindicating Friedman's critique of the original, purely statistical Phillips curve.

(ii) The Taylor Rule — the reasoning: a simple, transparent rule prescribing how a central bank's policy rate should respond to observable economic conditions: $i_{ff} = \pi + r^* + \frac{1}{2}(\pi - \pi^*) + \frac{1}{2}(y - y^*)$, where r^* is the equilibrium real fed funds rate, π^* is the inflation target, and the output gap measures how far real GDP is from potential.

Input	Value
Equilibrium real fed funds rate, r^*	2%
Inflation target, π^*	2%
Actual inflation, π	3%
Inflation gap ($\pi - \pi^*$)	1%
Output gap	1%
Taylor rule $i_{ff} = \pi + r^* + \frac{1}{2}(\text{inflation gap}) + \frac{1}{2}(\text{output gap}) = 3 + 2 + 0.5(1) + 0.5(1)$	
	6.0%

Why both gaps appear: including *both* an inflation gap and an output gap reflects that a central bank cares not only about keeping inflation near target, but also about smoothing business-cycle fluctuations of output around its potential — the Taylor rule is a compact summary of a central bank pursuing **both** objectives simultaneously, rather than inflation alone, and its transparency and simplicity are precisely why it's widely used as a benchmark for assessing whether actual policy looks "too loose" or "too tight" relative to what observable conditions would prescribe.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
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23-30	23-30	Accelerationist Phillips curve correctly derived from adaptive expectations with the $\pi_t = \pi_{t-1} - \gamma(U_t - U^*)$ relationship stated AND the Taylor rule formula correctly stated with a worked numeric example AND the reasoning behind each (why "accelerationist"; why both gaps in the Taylor rule) is explained.
15-22	15-22	Both correctly stated with formulas, but the underlying reasoning (adaptive expectations driving acceleration; dual mandate driving both Taylor rule gaps) is not explained.
8-14	8-14	Only one of the two correctly explained.
0-7	0-7	Neither correctly explained.

Classical Theory of Money Demand (Fisher's Quantity Theory): built from the equation of exchange, $MV = PY$. Fisher assumed **velocity (V) is fairly constant** in the short run (a matter of fixed institutional payment habits) and that output (Y) sits at its full-employment level given flexible wages and prices. Money demand is then $M_d = (1/V)PY = kPY$ — a fixed fraction of nominal income, with **no role for interest rates** at all. This makes the theory a clean theory of inflation: with V and Y essentially fixed, $\% \Delta M \approx \% \Delta P$.

Keynesian Liquidity Preference Theory: Keynes identified three motives for holding money — **transactions** and **precautionary** motives (both roughly proportional to income, as in the classical view) and, crucially, a **speculative** motive: money held as a store of wealth, as an alternative to holding bonds, when expected bond returns look unattractive. This makes money demand $M_d/P = f(i, Y)$ — negatively related to the interest rate as well as positively related to income — which means **velocity is not constant**: it moves procyclically with interest rates and with changing expectations about future "normal" interest rate levels, undermining the classical link between the money supply and nominal spending.

Friedman's Modern Quantity Theory: Friedman treated money as just another asset in a portfolio-choice framework, with demand for *real* money balances driven by **permanent income** (a longer-run, smoothed measure of wealth, rather than current income) and the expected returns on the alternative assets households could hold instead — bonds, equities, and physical goods (via expected inflation) — each relative to the return on money itself. Friedman argued that because these assets are not close substitutes for money in most portfolios, money demand is only **weakly sensitive to interest rates** — closer to the classical view in its practical stability, but grounded in explicit portfolio-choice reasoning rather than a fixed institutional velocity assumption. This is why Friedman's theory implies velocity is *not* literally constant (as Fisher assumed) but is **stable and predictable** — a crucial practical distinction from the Keynesian view, where velocity's sensitivity to volatile interest-rate expectations makes it genuinely unpredictable.

Comparing Money Demand Theories

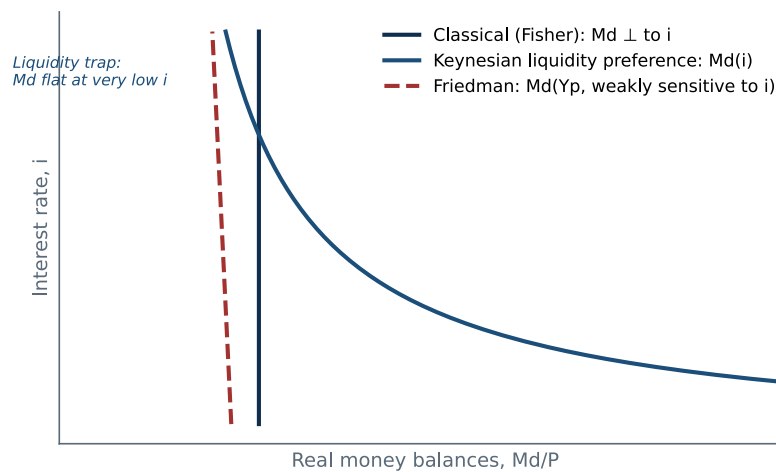


Fig. 2 — Comparing the three theories: Classical money demand is vertical (interest-rate-independent); Keynesian liquidity preference slopes down, including a flat "liquidity trap" region at very low rates; Friedman's is only weakly interest-sensitive, positioned by permanent income rather than current income.

	Classical (Fisher)	Keynesian (liquidity preference)	Friedman (modern quantity theory)
Interest-rate sensitivity	None	Strong (negative)	Weak
Income measure	Current nominal income	Current income	Permanent income
Velocity	Constant	Unstable/unpredictable	Stable, though not constant
Policy implication	Money supply tightly linked to nominal spending	Money supply loosely, unreliably linked to spending	Money supply reliably, though not mechanically, linked to spending

Opinion: which theory best fits the current economic environment?

Assessment: the most defensible position is that **Friedman's modern quantity theory offers the more useful practical framework today**, but not because interest rates are irrelevant — rather because the Keynesian liquidity preference framework's core policy implication (that velocity is so unstable the money supply is barely worth tracking) is less useful in an environment where most major central banks, including the Fed and ECB, explicitly target an **interest rate** (not a monetary aggregate) as their primary instrument — which is itself closer to what the Keynesian view of unstable velocity would recommend as sound practice (see Q. A1(c) and A2(a)) than to the classical view's implicit call to track M directly. In that sense, elements of *both* Keynesian and Friedman's reasoning show up in how central banks actually operate: interest-rate targeting (the Keynesian-consistent practical response to unstable velocity) implemented with an eye toward the same permanent-income-driven, portfolio-based stability Friedman emphasised, rather than the classical view's simple, mechanical $MV=PY$ link, which fits poorly with a world of quantitative easing (see Part (c)) where enormous expansions of the monetary base did not translate one-for-one into proportional inflation.

EXAM TECHNIQUE

This is a genuinely open question — a well-reasoned case for the Keynesian view (given the demonstrated importance of interest-rate expectations and the post-

2008/post-2020 liquidity-trap-like conditions in several major economies) or a case for the classical view (in a high-inflation, money-growth-driven environment) can score equally well. What earns marks is a reasoned, evidence-based argument connecting the theory's own assumptions to actually observed conditions, not simply asserting a preference.

Mark Allocation — Part (b) (50 marks)

Tier	Marks	What separates this tier
38-50	38-50	All three theories correctly explained with their core mechanisms (constant velocity/no interest sensitivity; three motives/interest-sensitive/unstable velocity; permanent income/weak interest sensitivity/stable velocity) AND a clear opinion given AND the opinion is justified by connecting the theory's own assumptions to specific, observable current conditions.
25-37	25-37	All three theories correctly explained, and an opinion given, but the justification is generic or doesn't engage with current conditions specifically.
13-24	13-24	Only two of the three theories correctly explained, or all three explained with no opinion/justification given at all.
0-12	0-12	Fewer than two theories correctly explained.

Conventional tools: the three tools central banks use in **normal times** to control the money supply and interest rates — **open market operations**, **discount lending**, and **reserve requirements** (see Q. A2(a)). All three work by adjusting the quantity of reserves in the banking system, and are well-understood, precise, and easily reversible.

Unconventional tools: measures developed specifically because, during the 2008 financial crisis, conventional tools proved insufficient once short-term interest rates hit the **zero lower bound** — there was no more room to cut a policy rate that was already near zero. Unconventional tools instead **directly target the cost and availability of external finance** to banks, households and firms — through central-bank liquidity provision, large-scale asset purchases (quantitative easing), and forward guidance (explicit communication about the likely future path of policy, intended to shape expectations directly rather than working solely through the current policy rate).

The key distinction: conventional tools work by moving the price of reserves (the fed funds rate) within a functioning market where that rate still has room to move; unconventional tools were developed for the situation where that room has run out, and instead act on the size and composition of the central bank's own balance sheet, or on market expectations directly, to keep providing monetary accommodation even at the zero lower bound.

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Conventional tools correctly named (OMO, discount lending, reserve requirements) AND unconventional tools correctly named (QE, forward guidance) AND the zero-lower-bound rationale for why unconventional tools were developed is explained.
10-14	10-14	Both categories of tool correctly named, but the zero-lower-bound rationale is missing.
5-9	5-9	Only one category correctly named.
0-4	0-4	No credible distinction given.

